

# Momentum, Reversal, Volatility and Size in U.S. Large Caps: A Cross-Sectional Factor Study with Fama-MacBeth Validation, 2011 to 2026

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## Abstract

This paper tests whether 12-1 momentum, one-month reversal, 60-day total volatility and log market capitalisation predict next-month returns among the 200 largest current S&P 500 constituents from February 2012 to August 2026, using decile sorts, Fama-MacBeth regressions with Newey-West standard errors and rank information coefficients. Momentum and volatility carry significant raw Fama-MacBeth premia ( $t = 2.91$  and  $3.07$ ), but risk adjustment separates them. The volatility long-short portfolio has a market beta of 1.17 and an insignificant CAPM alpha; its large five-factor-plus-momentum alpha (15.95% a year) comes from unprofitable, heavily investing firms that nonetheless won, the profile of firms selected by an end-of-sample universe. Momentum's market-adjusted premium of 3.7% a year per standard deviation ( $t = 3.41$ ) stays significant before 2023 and under point-in-time index membership ( $t = 2.26$ ), though its unadjusted point-in-time premium does not ( $t = 1.76$ ). It is the standard momentum factor, and its decile portfolio lost 56.3% from 2016 to 2023. The point-in-time filter removes the significance of a residual volatility premium and shrinks the negative size premium by a quarter, supporting a survivorship explanation. Reversal does not survive costs.

## 1 Introduction

Cross-sectional factor research asks a simple question: do stocks that share a measurable characteristic today earn systematically different returns next month? The question underpins most systematic equity strategies, and the standard tools for answering it are portfolio sorts and the two-step regression procedure of Fama and MacBeth (1973). This paper applies both, alongside rank information coefficients, to four characteristics with a long empirical history.

**Momentum.** Jegadeesh and Titman (1993) document that stocks with high returns over the prior three to twelve months continue to outperform. The standard 12-1 construction skips the most recent month to separate momentum from short-term reversal (Carhart, 1997). Daniel and Moskowitz (2016) show that momentum is prone to sharp crashes when markets rebound after a decline.

**Short-term reversal.** Jegadeesh (1990) and Lehmann (1990) find that short-horizon winners tend to underperform subsequently, an effect usually attributed to liquidity provision and concentrated in smaller, less liquid stocks.

**Volatility.** The volatility factor in this paper is total realised volatility over 60 trading days, which mixes market-related and stock-specific variation. Ang et al. (2006) report that stocks with high idiosyncratic volatility earn very low subsequent returns, and document the same pattern for total volatility; idiosyncratic volatility relative to a factor model is not constructed here. Baker

et al. (2011) and Frazzini and Pedersen (2014) relate the broader low-risk anomaly to leverage constraints and benchmarking. A positive volatility premium would therefore contradict the classic anomaly and requires explanation.

**Size.** Banz (1981) and Fama and French (1992) find that small firms earn higher average returns. Here size mainly serves as a control variable, and Section 6 explains why its estimated premium in this universe cannot be interpreted at face value.

The paper makes three points. First, the three evaluation lenses disagree in informative ways that locate each factor’s predictive content in the cross-section. Second, raw premia can mislead in a strong bull market: the most significant raw result, volatility, is largely market beta under the CAPM and looks like sample selection under a multi-factor model, while momentum only becomes clearly robust once market exposure is removed. Third, some results are products of the research design (survivorship bias from selecting the universe at the end of the sample), and a point-in-time membership filter tests part of this diagnosis directly. Given the problems of multiple testing (Harvey et al., 2016) and post-publication decay (McLean and Pontiff, 2016), interpretation is conservative throughout.

Four robustness extensions (drawdowns, multi-factor alphas, the point-in-time membership filter and non-winsorised factors) were added after the original results had been seen. Their specifications, including the rule for interpreting the membership filter, were fixed and committed to the project repository before any of them was computed, and none of them changes an original estimate.

## 2 Data

**Universe.** The universe is the 200 largest constituents of the S&P 500 by market capitalisation as of September 2026, taken from the current Wikipedia constituent list with GICS sector labels, and ranked using Yahoo Finance market capitalisations (from \$58.6 billion to \$5.17 trillion). The universe spans all 11 GICS sectors, from 39 Information Technology stocks to 5 Utilities. Because membership is fixed at the end of the sample, the universe is survivorship-biased by construction; the consequences are quantified in Section 6.

**Prices.** Daily dividend- and split-adjusted closing prices were downloaded from Yahoo Finance for 3,949 trading days from 3 January 2011 to 16 September 2026. Month-end prices are the last available close in each month. Because the price series ends mid-month, the incomplete September 2026 return is excluded, so the return sample ends in August 2026; the sample in which all four factors are available runs from February 2012 to August 2026 (175 months). Nineteen of the 200 stocks have less than 90% daily coverage because they listed or were spun off during the sample (for example AbbVie and Vertiv). A stock enters a factor only once it has enough history, so these late entrants account for 1,806 missing momentum observations, 5.2% of the momentum stock-month panel.

**Market data.** The market benchmark is the SPDR S&P 500 ETF (SPY), using dividend-adjusted prices as a total-return proxy, and the risk-free rate is the 13-week U.S. Treasury bill yield, both from Yahoo Finance. SPY holds the index constituents as they were at each point in time, so unlike an average of this paper’s universe it is free of the universe’s survivorship bias.

**Factor and membership data.** Monthly returns on the Fama-French five factors and the momentum factor (UMD) were downloaded from the Kenneth French Data Library on 2 October 2026; they are built from the full CRSP universe and run to August 2026. Historical S&P 500 membership comes from the dated constituent lists of the open-source `fja05680/sp500` repository (pinned to one commit, with lists to 18 August 2026). These lists record the ticker in use on each date, so eleven companies that changed symbol during the sample (for example Facebook to Meta) and one share class (Alphabet class C) are linked to their earlier symbols by an explicit,

date-bounded mapping, built by auditing every stock whose membership began after its price history.

**Data validation.** Before any factor was built, the price panel was checked for empty series, internal gaps and single-day moves above 50%. No series has internal gaps. Seven extreme daily moves were identified and six were verified against real events (for example the March 2020 oil price collapse); none was treated as an error. Table 1 summarises the resulting factor panels.

**Table 1:** Factor panels. Pooled statistics across all stock-months after winsorisation (size is not winsorised). Momentum and reversal are simple returns, volatility is 60-day total realised volatility, annualised, size is the natural log of market capitalisation in U.S. dollars.

| Factor           | First month | Months | Avg. stocks | Mean   | Std. dev. | P1     | P99    |
|------------------|-------------|--------|-------------|--------|-----------|--------|--------|
| Momentum         | 2012-02     | 175    | 189         | 0.192  | 0.362     | -0.412 | 1.350  |
| Reversal         | 2011-03     | 186    | 190         | 0.016  | 0.082     | -0.182 | 0.246  |
| Total volatility | 2011-04     | 185    | 190         | 0.279  | 0.144     | 0.108  | 0.832  |
| Size             | 2011-02     | 187    | 190         | 24.611 | 1.162     | 22.064 | 28.036 |

## 3 Methodology

### 3.1 Timing convention

Let  $P_{i,t}$  be the adjusted month-end price of stock  $i$  in month  $t$ . The return to be predicted is the return earned during month  $t$ ,

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1. \quad (1)$$

Every factor value paired with  $R_{i,t}$  uses only information available at the close of month  $t - 1$ . This single convention is applied to all factors and all tests, and it is the main safeguard against look-ahead bias. It was verified by recomputing individual factor values and returns by hand for sample stocks and months.

### 3.2 Factor definitions

$$\text{MOM}_{i,t} = \frac{P_{i,t-2}}{P_{i,t-13}} - 1 \quad (12\text{-}1 \text{ momentum: months } t - 12 \text{ to } t - 2) \quad (2)$$

$$\text{REV}_{i,t} = \frac{P_{i,t-1}}{P_{i,t-2}} - 1 \quad (\text{short-term reversal: month } t - 1) \quad (3)$$

$$\text{VOL}_{i,t} = \sqrt{252} \text{ sd}(r_{i,d})_{d \in \mathcal{D}_{t-1}} \quad (\text{total realised volatility}) \quad (4)$$

$$\text{SIZE}_{i,t} = \ln(S_i P_{i,t-1}) \quad (\log \text{ market capitalisation}) \quad (5)$$

Here  $r_{i,d}$  are daily simple returns and  $\mathcal{D}_{t-1}$  is the 60 trading days ending on the last trading day of month  $t - 1$ . VOL is total volatility: it is not purged of market or other factor exposure. The momentum and reversal windows do not overlap, so the two factors measure distinct horizons. Historical shares outstanding are not available from the data source, so market capitalisation uses the current share count  $S_i$  throughout, which ignores buybacks and issuance.

### 3.3 Winsorisation

Momentum, reversal and volatility are winsorised cross-sectionally each month at the 1st and 99th percentiles, which alters 2.11% of observations for each factor. Winsorising each month rather than against a single global cut-off keeps the thresholds appropriate for the volatility regime of

that month. The upper tail contained genuine but extreme values (for example a 12-1 momentum reading of more than 5,000% for SanDisk in 2026); the lower tail contained artificially low values (Vertiv’s near-constant price during its pre-merger SPAC period implied almost zero volatility). Size is not winsorised because the log transform already compresses its scale. Returns are never winsorised, so portfolio performance is not artificially smoothed.

### 3.4 Decile portfolios and turnover

Each month, stocks are ranked on each factor and assigned to deciles using the cross-sectional percentile rank  $u_{i,t} \in (0, 1]$ , with decile  $\lceil 10 u_{i,t} \rceil$ . Months with fewer than 100 valid stocks are excluded. Each decile is equal-weighted and held for one month. The long-short portfolio is always decile 10 minus decile 1 (high minus low), so that a negative spread means the factor is negatively related to future returns, matching the sign of the regression coefficients.

Long-short weights are  $w_{i,t} = 1/n_t^{10}$  for decile 10 stocks,  $-1/n_t^1$  for decile 1 stocks and zero otherwise. Turnover is

$$\text{TO}_t = \sum_i |w_{i,t} - w_{i,t-1}|, \quad (6)$$

which ranges from 0 (no change) to 4 (both legs entirely replaced). Weights are reset to equal each month, so drift in weights within the month is ignored.

### 3.5 Fama-MacBeth regressions

In the first step, for each month  $t$  with at least 100 stocks with complete data, the cross-section of returns is regressed on the factors:

$$R_{i,t} = a_t + \sum_k \gamma_{k,t} z_{k,i,t} + \varepsilon_{i,t}, \quad (7)$$

where  $z_{k,i,t}$  is factor  $k$  standardised to zero mean and unit standard deviation across the stocks in that month’s regression. Standardisation makes  $\gamma_{k,t}$  the return premium, in that month, for a one standard deviation higher exposure to factor  $k$ , comparable across factors and over time. Factors are estimated both one at a time (univariate) and jointly (multivariate), where each slope measures the premium holding the other factors fixed.

In the second step, the premium estimate is the time-series mean  $\bar{\gamma}_k = T^{-1} \sum_t \gamma_{k,t}$ . The classic Fama-MacBeth standard error,  $\text{sd}(\gamma_{k,t})/\sqrt{T}$ , assumes the monthly slopes are serially uncorrelated. Slowly moving characteristics such as momentum and size can make consecutive slopes correlated, so inference uses Newey-West heteroskedasticity and autocorrelation consistent standard errors (Newey and West, 1987) with a Bartlett kernel and  $L = \lfloor 4(T/100)^{2/9} \rfloor = 4$  lags (Newey and West, 1994). Plain Fama-MacBeth  $t$ -statistics are reported alongside for comparison. As a check on the implementation, the monthly slopes were recomputed independently with a direct least-squares solve and matched exactly.

### 3.6 Market risk adjustment

A long-short spread can be positive simply because its long leg has a higher market beta than its short leg, which pays off whenever the market rises. Two complementary adjustments address this. First, each long-short return series is regressed on the market excess return,

$$\text{LS}_t = \alpha + \beta (R_{m,t} - r_{f,t}) + e_t, \quad (8)$$

where  $R_{m,t}$  is the SPY return and  $r_{f,t}$  is the Treasury bill yield at the end of month  $t - 1$  divided by 12. Because a long-short portfolio is self-financing, its return is already an excess return. The

CAPM alpha  $\alpha$  (Sharpe, 1964) is the return not explained by market exposure; it is annualised and tested with Newey-West standard errors.

Second, each stock’s market beta is added to the Fama-MacBeth regressions as a fifth characteristic. Beta is the slope of the stock’s daily returns on SPY’s daily returns over the trailing 252 trading days, measured at the end of month  $t - 1$  and winsorised like the other factors. Controlling for beta as a characteristic is not the same as hedging the market, because each monthly slope  $\gamma_{k,t}$  is itself the return of a factor-mimicking portfolio that can still load on the market. The monthly slope series are therefore also regressed on the market excess return, which gives a market-adjusted premium for each factor.

### 3.7 Information coefficient

The information coefficient (IC) for factor  $k$  in month  $t$  is the Spearman rank correlation across stocks between  $f_{k,i,t}$  and  $R_{i,t}$  (Grinold and Kahn, 2000). The IC measures how well a factor orders stocks, uses every stock equally and is insensitive to outliers. The annualised information ratio of the IC is  $\text{ICIR} = \sqrt{12}\text{IC}/\text{sd}(\text{IC})$ .

### 3.8 Validation design

**Out-of-sample split.** The 175-month sample is split at one calendar date for every factor: the first 60% (105 months, to October 2020) is in-sample and the remaining 70 months are out-of-sample. Because no parameter is estimated in-sample (all factor definitions are fixed in advance from the literature), the split tests the stability of the premia over time rather than overfitting in the usual sense.

**Transaction costs.** Net returns deduct  $c \cdot \text{TO}_t$  each month with  $c = 10$  basis points per unit of turnover, and the first month is charged the full initial build ( $\text{TO} = 2$ ). Sensitivity to  $c \in \{0, 5, 10, 20\}$  basis points and the break-even cost  $\bar{R}^{\text{gross}}/\overline{\text{TO}}$  are reported.

**Sector neutrality.** A sector-neutral version ranks each stock within its GICS sector before forming deciles, and adds sector dummies to the regressions (Section 5.6).

## 4 Results

The size results in this section are reported for completeness; Section 6 explains why they reflect the construction of the universe rather than a size effect.

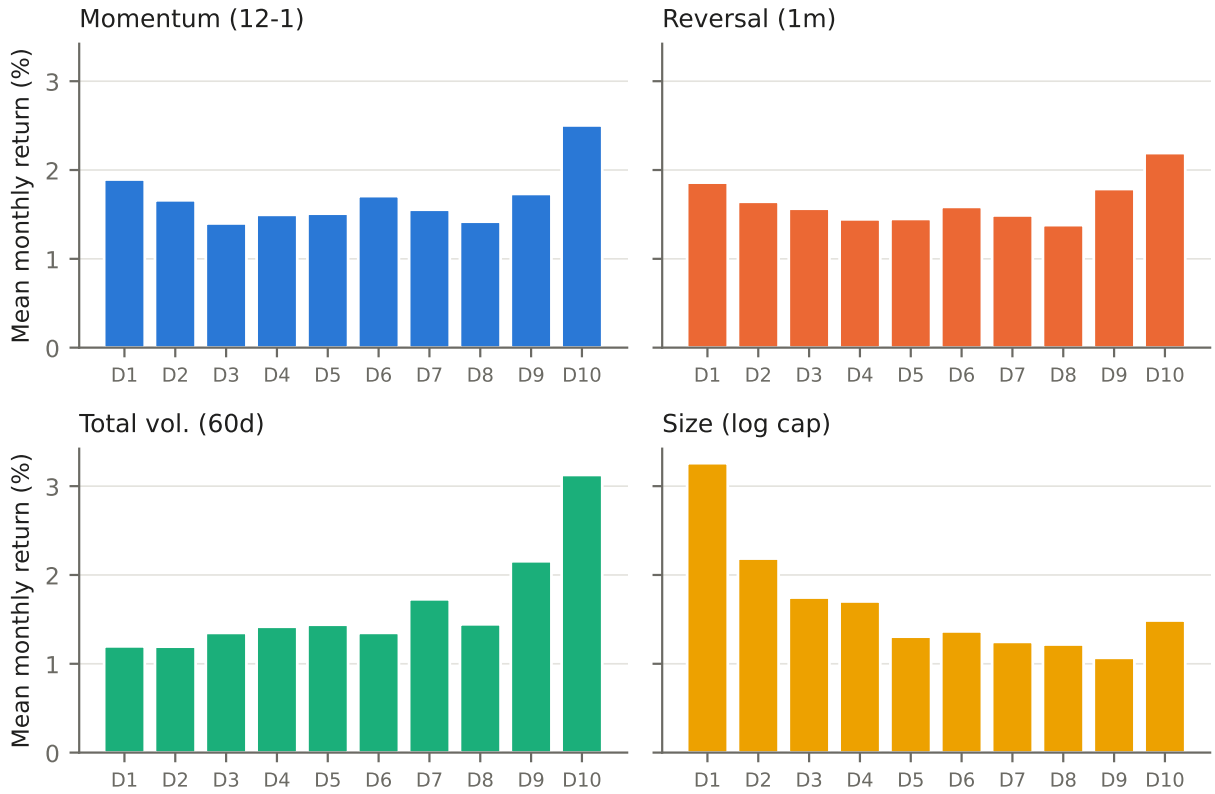
### 4.1 Decile portfolios

Table 2 and Figure 1 show mean monthly returns by decile. The volatility sort is close to monotonic, with decile 10 earning 3.13% per month against 1.20% for decile 1. The momentum and reversal sorts are U-shaped: both extreme deciles outperform the middle of the distribution. Stocks with extreme past returns, whether winners or losers, tend to be high-volatility stocks (average volatility is 0.36 in both momentum decile 1 and decile 10, against about 0.24 in deciles 4 to 7), so the extreme momentum and reversal deciles partly inherit the volatility effect. A linear rank correlation between momentum and volatility is close to zero, because it cannot see a U-shaped relationship.

Table 3 reports the long-short spreads. Only volatility (Newey-West  $t = 2.83$ ) and size ( $t = -4.92$ ) are significant. The momentum spread earns 7.31% a year gross with a Sharpe ratio of 0.29 and is not significant ( $t = 1.18$ ): the short leg of losers performs well and erodes the spread. Figure 2 shows the cumulative performance of each spread.

**Table 2:** Mean monthly return (%) of equal-weighted decile portfolios. D1 holds the lowest factor values, D10 the highest.

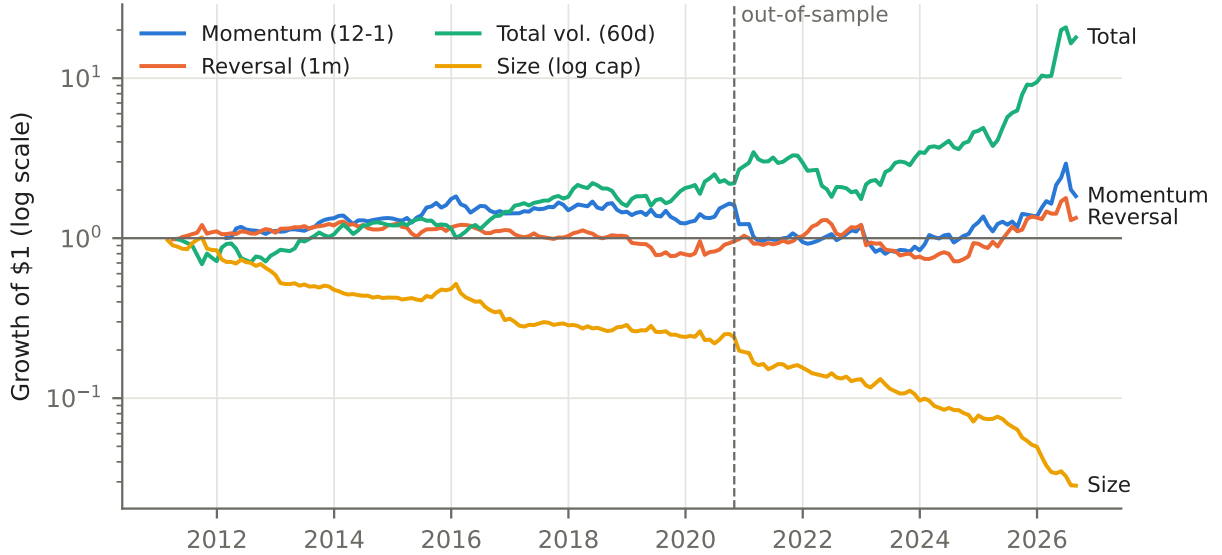
| Factor           | D1   | D2   | D3   | D4   | D5   | D6   | D7   | D8   | D9   | D10  |
|------------------|------|------|------|------|------|------|------|------|------|------|
| Momentum         | 1.89 | 1.66 | 1.40 | 1.50 | 1.51 | 1.71 | 1.55 | 1.42 | 1.73 | 2.50 |
| Reversal         | 1.86 | 1.64 | 1.56 | 1.44 | 1.45 | 1.59 | 1.49 | 1.38 | 1.79 | 2.19 |
| Total volatility | 1.20 | 1.19 | 1.35 | 1.42 | 1.44 | 1.35 | 1.73 | 1.44 | 2.15 | 3.13 |
| Size             | 3.26 | 2.18 | 1.75 | 1.70 | 1.31 | 1.36 | 1.25 | 1.22 | 1.07 | 1.49 |



**Figure 1:** Mean monthly return by decile. Volatility is close to monotonic; momentum and reversal are U-shaped; size falls from D1 to D9.

**Table 3:** Decile 10 minus decile 1 long-short portfolios, gross of costs. Newey-West  $t$ -statistic on the mean monthly return with 4 lags. Turnover is the mean monthly  $TO_t$  (maximum 4).

| Factor           | Months | Mean (%/mo) | Ann. return (%) | Ann. vol (%) | Sharpe | NW $t$ | Turnover |
|------------------|--------|-------------|-----------------|--------------|--------|--------|----------|
| Momentum         | 175    | 0.61        | 7.31            | 24.93        | 0.29   | 1.18   | 1.09     |
| Reversal         | 186    | 0.33        | 3.99            | 20.25        | 0.20   | 0.87   | 3.44     |
| Total volatility | 185    | 1.93        | 23.16           | 30.14        | 0.77   | 2.83   | 1.00     |
| Size             | 187    | -1.77       | -21.26          | 16.37        | -1.30  | -4.92  | 0.22     |



**Figure 2:** Growth of \$1 in each long-short portfolio (gross, log scale). The dashed line marks the start of the out-of-sample period (November 2020).

## 4.2 Fama-MacBeth regressions

Table 4 reports the regression results. In the multivariate specification, momentum earns 0.30% per month per standard deviation ( $t = 2.91$ ) and volatility 0.49% ( $t = 3.07$ ); reversal is insignificant ( $t = 0.51$ ) and size is negative ( $t = -3.85$ ). The monthly regressions explain on average 14.5% of cross-sectional return variation across an average of 189 stocks.

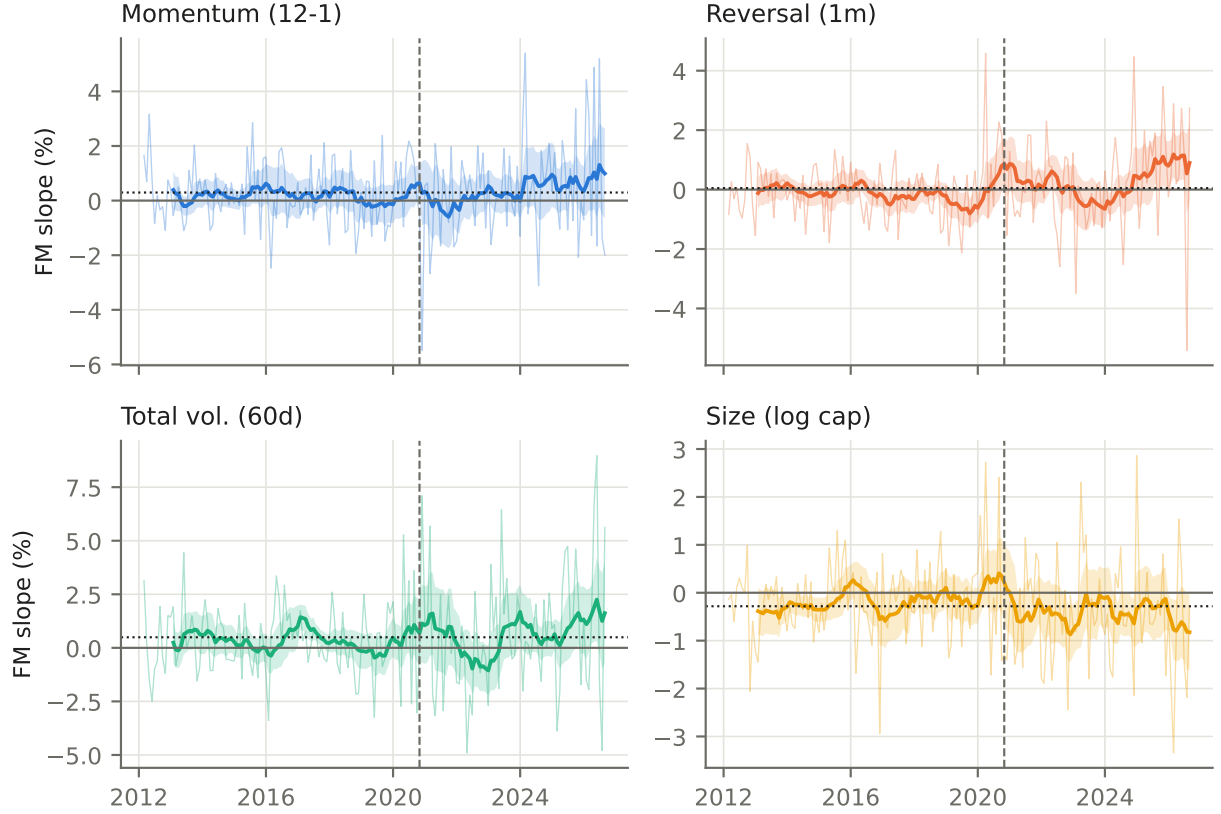
Three features of the table deserve comment. First, momentum is significant in the regression but not in the decile sort: the regression uses every stock with a linear weight, while the sort uses only the two extreme deciles, where the U-shape works against the spread. Second, the momentum premium barely changes between the univariate and multivariate specifications (0.285% and 0.296%); its  $t$ -statistic rises because the other factors absorb cross-sectional noise. Third, Newey-West and plain  $t$ -statistics are similar, so the monthly slopes are only modestly autocorrelated. Figure 3 plots the slopes over time.

**Table 4:** Fama-MacBeth regressions of monthly returns on standardised factors.  $\bar{\gamma}$  is the mean monthly slope in % per one cross-sectional standard deviation. Newey-West  $t$ -statistics (4 lags) in parentheses. Multivariate sample: 175 months, February 2012 to August 2026.

| Factor           | Univariate         |         | Multivariate       |         |              |              |
|------------------|--------------------|---------|--------------------|---------|--------------|--------------|
|                  | $\bar{\gamma}$ (%) | NW $t$  | $\bar{\gamma}$ (%) | NW $t$  | Plain FM $t$ | % months > 0 |
| Momentum         | 0.285              | (2.01)  | 0.296              | (2.91)  | (2.63)       | 56.6         |
| Reversal         | 0.096              | (0.84)  | 0.050              | (0.51)  | (0.53)       | 52.0         |
| Total volatility | 0.599              | (3.14)  | 0.493              | (3.07)  | (3.00)       | 59.4         |
| Size             | -0.443             | (-5.35) | -0.282             | (-3.85) | (-3.92)      | 37.1         |
| Intercept        |                    |         | 1.686              | (6.68)  | (5.19)       | 72.6         |

## 4.3 Information coefficients

Table 5 reports rank ICs. Only size has a significant mean IC. The mean ICs for momentum (0.010) and volatility (0.028) are small and insignificant, and momentum's IC is positive in only 50.9% of



**Figure 3:** Monthly multivariate Fama-MacBeth slopes (faint), their 12-month rolling mean (solid) with a 95% band, and the full-sample mean (dotted). The dashed line marks the out-of-sample split.

months. Figure 4 shows the IC over time.

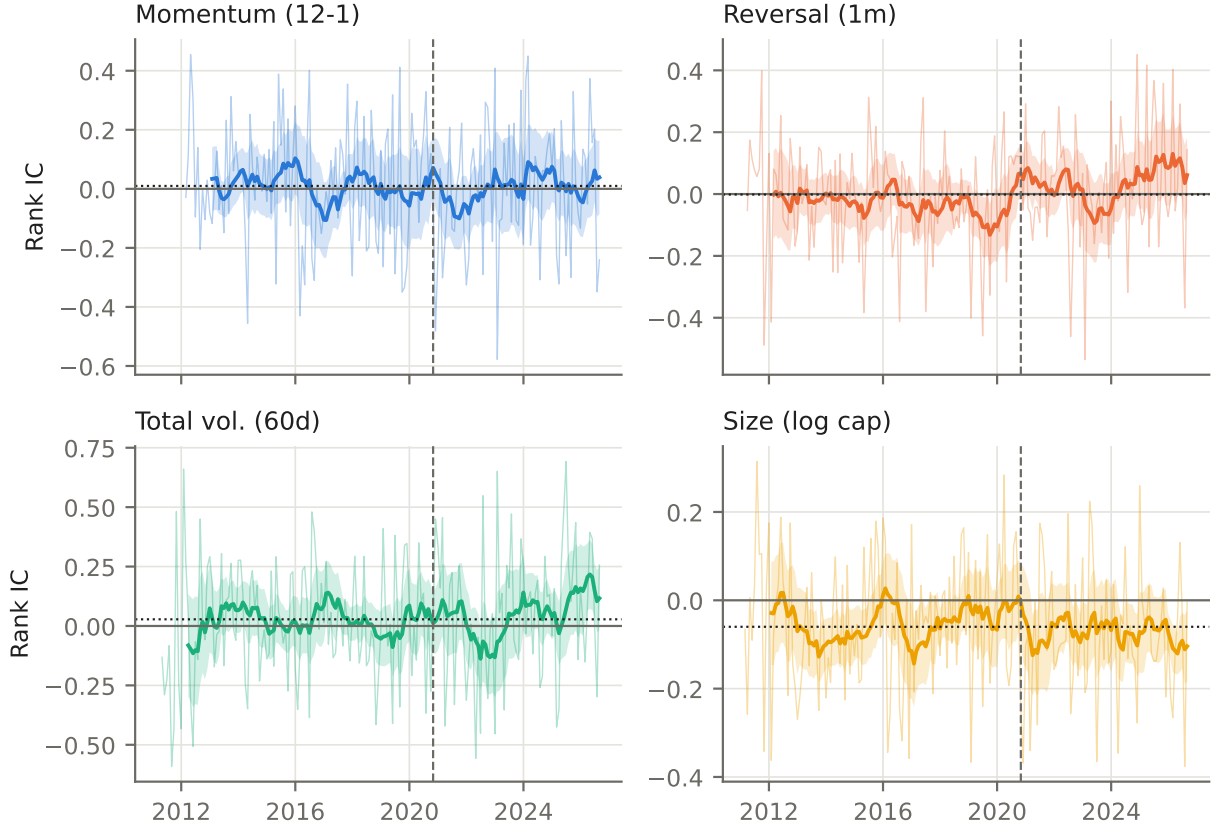
**Table 5:** Monthly Spearman rank information coefficients between factor values and next-month returns. Newey-West  $t$  on the mean IC (4 lags).

| Factor           | Months | Mean IC | IC std. dev. | ICIR (ann.) | NW $t$ | % months IC > 0 |
|------------------|--------|---------|--------------|-------------|--------|-----------------|
| Momentum         | 175    | 0.010   | 0.209        | 0.16        | 0.69   | 50.9            |
| Reversal         | 186    | -0.002  | 0.180        | -0.03       | -0.12  | 47.8            |
| Total volatility | 185    | 0.028   | 0.256        | 0.38        | 1.55   | 54.1            |
| Size             | 187    | -0.060  | 0.145        | -1.43       | -5.99  | 33.7            |

#### 4.4 Reconciling the three lenses

The lenses answer different questions. The IC asks whether a factor orders all stocks correctly, giving each stock equal weight. The regression asks whether return is linearly related to the factor's magnitude, so extreme observations carry more weight. The decile spread asks whether the extreme deciles differ. For momentum and volatility the regression is significant while the IC is not, so the predictive content is concentrated in the tails of the cross-section (a handful of extreme stocks) rather than spread evenly across all 200 names. Section 5 examines whether these premia are stable over time and whether they survive adjustment for market exposure.





**Figure 4:** Monthly rank IC (faint) and its 12-month rolling mean with a 95% band.

## 5 Robustness and Validation

### 5.1 Out-of-sample and subperiod stability

In the 60/40 split, momentum’s Fama-MacBeth premium is significant in-sample ( $t = 2.46$ ) but not out-of-sample ( $t = 1.80$ ). Volatility is significant in both halves, but its long-short return rises from 12.61% to 43.68% a year, which is not a plausible stationary premium.

A split at the end of 2022 is more revealing (Table 6). Before 2023, neither raw premium is significant at the 5% level (momentum  $t = 1.87$ , volatility  $t = 1.56$ ). From 2023 to August 2026, a period of 44 months dominated by the rally in AI, memory and semiconductor stocks, both raw premia are three to five times larger, and the volatility long-short portfolio earned 6.06% per month against 0.64% before 2023. Two explanations compete for this concentration: a genuine market episode, and the fact that the universe is selected on September 2026 market capitalisation, so that selection is strongest for stocks whose value rose most in the final years (SanDisk, for example, is in the universe only because of its 2025 to 2026 rise). Section 5.2 shows that once market exposure is removed, the picture differs sharply between the two factors.

**Table 6:** Multivariate Fama-MacBeth premia before and during the 2023 to 2026 rally.

| Factor           | 2012-02 to 2022-12 (131 months) |         | 2023-01 to 2026-08 (44 months) |         |
|------------------|---------------------------------|---------|--------------------------------|---------|
|                  | $\bar{\gamma}$ (%)              | NW $t$  | $\bar{\gamma}$ (%)             | NW $t$  |
| Momentum         | 0.191                           | (1.87)  | 0.608                          | (2.48)  |
| Reversal         | -0.028                          | (-0.30) | 0.283                          | (1.14)  |
| Total volatility | 0.251                           | (1.56)  | 1.216                          | (3.28)  |
| Size             | -0.237                          | (-2.94) | -0.414                         | (-2.52) |

Momentum’s worst months are listed in Table 7. Two of them, November 2020 and February 2021, came in the recovery from the 2020 crash, when past losers outperformed past winners: the rebound pattern described by Daniel and Moskowitz (2016). The worst month, July 2026, was a sharp reversal within the semiconductor rally itself: the winner decile, led by SanDisk, Corning, KLA, Marvell and Intel, fell 20.9% while the loser decile rose 10.4%.

**Table 7:** The five worst monthly returns of the momentum long-short portfolio.

| Month   | Momentum L/S return (%) |
|---------|-------------------------|
| 2026-07 | −31.4                   |
| 2020-11 | −24.3                   |
| 2023-01 | −19.8                   |
| 2021-02 | −16.8                   |
| 2025-02 | −11.7                   |

## 5.2 Market risk adjustment

Table 8 reports CAPM regressions of the long-short portfolios on the SPY excess return. The volatility spread is a leveraged market position: its beta is 1.17 ( $t = 6.99$ ) and the regression explains 30% of its variance. Its CAPM alpha of 8.30% a year is not significant ( $t = 1.31$ ) and is negative before 2023 (−3.13%,  $t = -0.54$ ); all of the remaining alpha comes from 2023 to 2026 (38.80% a year). Within the single-factor model, the volatility spread is therefore market exposure (Section 5.3 gives a different reading). Momentum behaves in the opposite way. Its beta is slightly negative over the full sample, its alpha of 8.97% a year is earned mainly before 2023, and its raw strength after 2023 came with a beta of 0.89, because the winner decile in that period consisted largely of high-beta technology stocks. Momentum’s decile alphas are only marginally significant ( $t$  between 1.46 and 1.71), consistent with the noise of 20-stock deciles and the U-shape discussed in Section 4.

**Table 8:** CAPM regressions of long-short portfolios on the SPY excess return. Beta is from the full-sample gross regression. Alphas are annualised with Newey-West  $t$ -statistics (4 lags) in parentheses. Net alphas deduct 10 basis points per unit turnover.

| Factor           | Market beta |         | CAPM alpha (%/yr), NW $t$ |                |                |                |
|------------------|-------------|---------|---------------------------|----------------|----------------|----------------|
|                  | $\beta$     | NW $t$  | Gross                     | Net            | 2012 to 2022   | 2023 to 2026   |
| Momentum         | −0.13       | (−0.69) | 8.97 (1.71)               | 7.65 (1.46)    | 8.16 (1.66)    | 3.97 (0.27)    |
| Reversal         | −0.32       | (−2.15) | 7.98 (1.77)               | 3.86 (0.85)    | 7.46 (1.73)    | 6.41 (0.55)    |
| Total volatility | 1.17        | (6.99)  | 8.30 (1.31)               | 7.11 (1.12)    | −3.13 (−0.54)  | 38.80 (2.74)   |
| Size             | −0.43       | (−4.26) | −15.76 (−3.79)            | −16.03 (−3.85) | −10.24 (−2.76) | −35.79 (−3.05) |

Table 9 turns to the regressions. Adding each stock’s beta as a fifth characteristic leaves the momentum and volatility premia significant ( $t = 3.15$  and  $3.58$ ), while beta itself earns nothing ( $t = 0.59$ ). The market-adjusted slope alphas give the clearest picture. Momentum’s premium is 3.68% a year per standard deviation ( $t = 3.41$ ), essentially unaffected by the beta control, and remains significant before 2023 ( $t = 2.49$ ). The volatility slope, by contrast, loads heavily on the market (slope beta 0.26,  $t = 6.38$  in the four-factor model); after market adjustment its premium is insignificant ( $t = 1.63$ ) and close to zero before 2023 ( $t = 0.28$ ).

One result needs particular care. Holding each stock’s beta fixed and hedging the market, volatility retains a positive premium (3.59% a year,  $t = 3.05$ ;  $t = 2.20$  before 2023), while beta’s own market-adjusted premium is negative (−4.22% a year before 2023,  $t = -2.62$ ). The negative beta premium is consistent with betting against beta (Frazzini and Pedersen, 2014), but the

**Table 9:** Fama-MacBeth regressions with trailing market beta as a control, and market-adjusted premia. The first two columns are mean monthly slopes from the five-characteristic model. The remaining columns are CAPM alphas of the monthly slope series (annualised, % per one standard deviation), from the four-factor model of Table 4 and from the model with the beta control.

| Factor           | FM with beta control |         | Market-adjusted slope alpha (%/yr), NW $t$ |               |                   |               |
|------------------|----------------------|---------|--------------------------------------------|---------------|-------------------|---------------|
|                  | $\bar{\gamma}$ (%)   | NW $t$  | Four-factor model                          |               | With beta control |               |
|                  |                      |         | Full                                       | 2012 to 2022  | Full              | 2012 to 2022  |
| Momentum         | 0.295                | (3.15)  | 3.68 (3.41)                                | 2.83 (2.49)   | 3.68 (3.62)       | 2.80 (2.69)   |
| Reversal         | 0.035                | (0.37)  | 1.28 (1.05)                                | 0.40 (0.33)   | 1.04 (0.86)       | 0.34 (0.27)   |
| Total volatility | 0.423                | (3.58)  | 2.52 (1.63)                                | 0.46 (0.28)   | 3.59 (3.05)       | 3.13 (2.20)   |
| Size             | -0.264               | (-3.57) | -3.03 (-3.03)                              | -2.44 (-2.31) | -2.69 (-2.66)     | -1.92 (-1.80) |
| Beta             | 0.096                | (0.59)  |                                            |               | -1.97 (-1.19)     | -4.22 (-2.62) |

positive premium on volatility in excess of beta contradicts Ang et al. (2006). The most plausible explanation is universe selection: a volatile stock is more likely than a stable one to rise far enough to enter the 2026 top 200. This residual premium is therefore treated as a symptom of survivorship bias, not a finding, and Section 5.7 tests that diagnosis.

### 5.3 Multi-factor risk adjustment

This subsection and Sections 5.5, 5.7 and 5.8 are the pre-specified robustness extensions. Each long-short return series and each multivariate Fama-MacBeth slope series is regressed on the same calendar month's returns of the CAPM (French market factor) and of the five-factor model of Fama and French (2015) (market, SMB, HML, profitability RMW and investment CMA) plus the momentum factor UMD. As an alignment check, SPY's excess return has a correlation of 0.995 with the French market factor (-0.13 when misaligned by one month), and the French CAPM alphas (Table 10) closely reproduce those of Table 8. Net of 10 basis points of costs, the alphas are about 1.2 to 1.3 percentage points a year lower for momentum and volatility. Table 11 reports the FF5+UMD loadings.

**Table 10:** Annualised alphas against the CAPM and the Fama-French five-factor model plus momentum (FF5+UMD), with Newey-West  $t$ -statistics in parentheses. French factor data run to August 2026, so every regression uses the full return sample.

| Series                                                           | Full sample    |                | FF5+UMD by period |                |
|------------------------------------------------------------------|----------------|----------------|-------------------|----------------|
|                                                                  | CAPM           | FF5+UMD        | 2012 to 2022      | 2023 to 2026   |
| <i>Panel A: long-short portfolios, gross (%/yr)</i>              |                |                |                   |                |
| Momentum                                                         | 9.01 (1.70)    | -0.71 (-0.24)  | -1.72 (-0.58)     | 10.00 (0.94)   |
| Reversal                                                         | 7.92 (1.76)    | 5.27 (1.15)    | 2.93 (0.69)       | 15.14 (1.38)   |
| Total volatility                                                 | 7.98 (1.28)    | 15.95 (3.51)   | 8.68 (2.16)       | 47.35 (3.82)   |
| Size                                                             | -15.42 (-3.75) | -20.22 (-6.46) | -15.58 (-5.98)    | -39.85 (-3.65) |
| <i>Panel B: multivariate Fama-MacBeth slopes (%/yr per 1 SD)</i> |                |                |                   |                |
| Momentum                                                         | 3.69 (3.41)    | 1.79 (2.10)    | 0.81 (1.01)       | 7.20 (2.15)    |
| Reversal                                                         | 1.25 (1.03)    | 0.60 (0.52)    | -0.83 (-0.74)     | 5.56 (2.15)    |
| Total volatility                                                 | 2.36 (1.56)    | 4.42 (3.39)    | 3.19 (2.32)       | 8.98 (3.08)    |
| Size                                                             | -2.97 (-2.97)  | -3.25 (-5.21)  | -2.53 (-3.98)     | -5.76 (-2.86)  |

**Momentum.** The momentum long-short portfolio loads 1.63 on UMD ( $t = 12.02$ ,  $R^2 = 0.70$ ), and its alpha disappears once UMD is included (-1.48% a year,  $t = -0.46$ , in the Carhart (1997) four-factor model; -0.71%,  $t = -0.24$ , in FF5+UMD). The Fama-MacBeth momentum slope also loads heavily on UMD ( $t = 8.83$ ). Its alpha beyond the factor models is 1.79% a year ( $t = 2.10$ ) over the full sample, but only 0.81% ( $t = 1.01$ ) before 2023. This is the expected result: 12-1

**Table 11:** Factor loadings in the FF5+UMD model, full sample (Newey-West  $t$ -statistics in parentheses). Long-short returns are monthly decimals; Fama-MacBeth slopes are in % per month per standard deviation.

| Series                              | Mkt-RF        | SMB           | HML           | RMW           | CMA           | UMD           | $R^2$ |
|-------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
| <i>Long-short portfolios, gross</i> |               |               |               |               |               |               |       |
| Momentum                            | 0.28 (3.05)   | -0.03 (-0.21) | 0.24 (2.19)   | -0.44 (-2.15) | -0.24 (-1.39) | 1.63 (12.02)  | 0.70  |
| Reversal                            | -0.11 (-0.84) | -0.28 (-1.32) | -0.16 (-0.95) | -0.56 (-1.48) | 0.75 (3.20)   | 0.40 (2.27)   | 0.23  |
| Total volatility                    | 0.90 (6.25)   | -0.04 (-0.18) | 0.09 (0.41)   | -1.75 (-5.82) | -1.08 (-3.80) | -0.29 (-1.77) | 0.65  |
| Size                                | -0.25 (-3.40) | -0.61 (-4.38) | -0.27 (-2.35) | 0.15 (0.84)   | 0.01 (0.03)   | 0.28 (3.36)   | 0.47  |
| <i>Fama-MacBeth slopes</i>          |               |               |               |               |               |               |       |
| Momentum                            | 0.06 (3.89)   | 0.04 (0.82)   | 0.01 (0.35)   | 0.05 (0.84)   | -0.06 (-1.24) | 0.31 (8.83)   | 0.53  |
| Reversal                            | -0.01 (-0.43) | -0.02 (-0.41) | -0.07 (-1.80) | -0.11 (-1.35) | 0.14 (2.48)   | 0.10 (3.53)   | 0.21  |
| Total volatility                    | 0.19 (4.79)   | -0.02 (-0.41) | -0.04 (-0.75) | -0.42 (-5.01) | -0.18 (-2.14) | -0.15 (-4.33) | 0.55  |
| Size                                | -0.02 (-1.20) | -0.13 (-4.56) | -0.09 (-4.19) | -0.04 (-1.54) | -0.06 (-1.44) | -0.01 (-0.31) | 0.41  |

momentum here is the standard momentum factor, with no evidence of an additional source of return outside 2023 to 2026.

**Volatility.** Multi-factor adjustment raises rather than lowers the volatility alpha. The long-short portfolio loads  $-1.75$  on RMW ( $t = -5.82$ ) and  $-1.08$  on CMA ( $t = -3.80$ ): its long leg holds unprofitable, aggressively investing firms, which earn low returns in the broad market, so the model predicts far less than the portfolio earned. The FF5+UMD alpha is 15.95% a year ( $t = 3.51$ ), and 8.68% ( $t = 2.16$ ) before 2023. In the full CRSP universe such exposures explain the low-volatility anomaly (Fama and French, 2016); here they leave a large positive alpha. The more economical explanation is selection: such a firm is among the 200 largest in 2026 only if its bets paid off, and the survivorship-free French factors cannot capture that. No specification supports a tradable volatility premium.

**Size and reversal.** The big-minus-small spread loads  $-0.61$  on SMB ( $t = -4.38$ ), as expected, yet its alpha stays strongly negative ( $-20.22\%$  a year,  $t = -6.46$ ): the true size factor explains little of it, which is consistent with a construction artefact. Reversal is insignificant in every model.

## 5.4 Transaction costs

Table 12 applies transaction costs. Short-term reversal replaces about 86% of each leg every month (turnover 3.44), which at 10 basis points per unit costs 4.13% a year, more than its 3.99% gross return; its break-even cost is 9.7 basis points, below realistic trading costs. Momentum and volatility have much lower turnover and break-even costs of 56.0 and 193.0 basis points, so costs reduce but do not eliminate their gross results. The cost model is deliberately simple (Section 6); Novy-Marx and Velikov (2016) show that more realistic cost models eliminate many published anomalies.

**Table 12:** Gross and net long-short performance. Cost is 10 basis points per unit turnover, with the initial portfolio build charged in the first month. Break-even is the cost per unit turnover at which the mean net return is zero (not defined for a negative gross spread).

| Factor           | Turnover | Cost (%/yr) | Gross ret. (%) | Net ret. (%) | Break-even (bps) | Net Sharpe at cost (bps per unit turnover) |       |       |       |
|------------------|----------|-------------|----------------|--------------|------------------|--------------------------------------------|-------|-------|-------|
|                  |          |             |                |              |                  | 0                                          | 5     | 10    | 20    |
| Momentum         | 1.09     | 1.30        | 7.31           | 6.00         | 56.0             | 0.29                                       | 0.27  | 0.24  | 0.19  |
| Reversal         | 3.44     | 4.13        | 3.99           | -0.13        | 9.7              | 0.20                                       | 0.10  | -0.01 | -0.21 |
| Total volatility | 1.00     | 1.20        | 23.16          | 21.95        | 193.0            | 0.77                                       | 0.75  | 0.73  | 0.69  |
| Size             | 0.22     | 0.26        | -21.26         | -21.53       | n/a              | -1.30                                      | -1.31 | -1.31 | -1.33 |

## 5.5 Drawdowns

Table 13 reports the maximum drawdown of each long-short portfolio, measured on wealth compounded from its monthly returns, with the peak, trough and recovery months, and the Calmar

ratio (annualised return divided by the absolute maximum drawdown).<sup>1</sup>

**Table 13:** Maximum drawdown (Max DD) of compounded monthly wealth and Calmar ratio (annualised arithmetic return divided by |Max DD|). Peak, trough and recovery months refer to the gross series; recovery is the first month in which wealth regains its pre-drawdown peak.

| Factor           | Gross         |            |         |         |               |        | Net of 10 bps |        |
|------------------|---------------|------------|---------|---------|---------------|--------|---------------|--------|
|                  | Ann. ret. (%) | Max DD (%) | Peak    | Trough  | Recovery      | Calmar | Max DD (%)    | Calmar |
| Momentum         | 7.31          | -56.3      | 2016-01 | 2023-05 | 2026-04       | 0.13   | -60.4         | 0.10   |
| Reversal         | 3.99          | -44.8      | 2022-05 | 2024-07 | 2025-10       | 0.09   | -65.4         | 0.00   |
| Total volatility | 23.16         | -49.1      | 2021-02 | 2022-12 | 2024-02       | 0.47   | -50.2         | 0.44   |
| Size             | -21.26        | -97.2      | 2011-09 | 2026-08 | not recovered | -0.22  | -97.3         | -0.22  |

Drawdowns are large relative to returns. The momentum portfolio lost 56.3% between January 2016 and May 2023 and regained its peak only in April 2026; with a Calmar ratio of 0.13, its worst drawdown equals nearly eight years of its average gross return. The positive pre-2023 CAPM alpha in Table 8 should be read alongside this: much of it reflects the portfolio’s negative market beta during a rising market, not steady positive returns. Volatility lost 49.1% from February 2021 to December 2022, through the 2022 bear market, as its market beta implies, and it has the highest Calmar ratio (0.47). Size lost 97.2% and never recovered, the compounded result of its negative mean. Costs deepen the drawdown of the high-turnover reversal portfolio most, from 44.8% to 65.4%, with no recovery. None of the portfolios would be investable on a stand-alone basis at the risk implied by equal-weighted 20-stock deciles.

## 5.6 Sector-neutral construction

A natural concern is that the volatility and momentum spreads are disguised sector bets, since the high-volatility decile in 2023 to 2026 is dominated by technology stocks. Ranking within sectors addresses this directly: the Information Technology share of the volatility top decile after January 2023 falls from 58% to 20%, and the volatility spread falls from 23.16% to 15.92% a year. Fama-MacBeth premia with sector dummies stay close to the originals (momentum 0.290%,  $t = 2.64$ ; volatility 0.420%,  $t = 2.87$ ). At the level of the 11 GICS sectors the factors are therefore not simply sector bets; finer sub-industry neutralisation was not tested.

## 5.7 Point-in-time index membership

This extension tests the survivorship diagnosis of the size and residual volatility premia directly. The universe in month  $t$  is restricted to the same 200 stocks, but only in months in which they were S&P 500 members at the end of month  $t - 1$  (Section 2), so no future information enters. Everything else is unchanged, with winsorisation re-applied to each month’s filtered cross-section.

The filter removes each stock from the months before it joined the index, the main channel of end-of-sample selection: Tesla, Palantir and SanDisk, for example, enter only after their index additions in December 2020, September 2024 and November 2025. It cannot restore firms that left the index, because their prices are not in the panel; prices for former constituents were deliberately not downloaded, since recycled tickers can silently attach a different company’s prices. The filtered universe averages 155 stocks a month in 2012, against 176, and converges to the main universe by 2026 (Table 14); no month falls below the 100-stock minimum.

The interpretation rule was fixed before the check was run. The size diagnosis is supported if the size spread shrinks in magnitude by at least half or its Fama-MacBeth  $t$ -statistic falls below 2 in absolute value, partially supported if the spread shrinks by 25% to 50%, and not confirmed

<sup>1</sup>Drawdowns are measured on month-end wealth, so a fall that opens and closes within a month is not captured; the figures understate drawdowns measured on daily data.

**Table 14:** Average number of stocks per month with all four factors and a return, by year, in the main and point-in-time universes.

| Year          | 2012 | 2014 | 2016 | 2018 | 2020 | 2022 | 2024 | 2026 |
|---------------|------|------|------|------|------|------|------|------|
| Main universe | 176  | 182  | 185  | 188  | 190  | 196  | 198  | 200  |
| Point-in-time | 155  | 162  | 168  | 176  | 179  | 180  | 188  | 198  |

otherwise. The residual volatility diagnosis is supported if the market-adjusted volatility premium in the beta-control model falls below  $t = 2$ .

**Table 15:** Main universe (fixed September 2026 top 200) against the point-in-time (PIT) universe (the same stocks, only while S&P 500 members at the end of the prior month). Market-adjusted premia are CAPM alphas of the monthly Fama-MacBeth slopes, as in Table 9. Sample: 175 months, February 2012 to August 2026, for the regressions.

|                                                                                                                     |                            |                |                              |               |                               |               |
|---------------------------------------------------------------------------------------------------------------------|----------------------------|----------------|------------------------------|---------------|-------------------------------|---------------|
| <i>Panel A: decile long-short portfolios, gross</i>                                                                 |                            |                |                              |               |                               |               |
|                                                                                                                     | Ann. return (%)            |                | Sharpe                       |               | NW $t$                        |               |
|                                                                                                                     | Main                       | PIT            | Main                         | PIT           | Main                          | PIT           |
| Momentum                                                                                                            | 7.31                       | 2.43           | 0.29                         | 0.10          | 1.18                          | 0.41          |
| Reversal                                                                                                            | 3.99                       | 0.90           | 0.20                         | 0.05          | 0.87                          | 0.22          |
| Total volatility                                                                                                    | 23.16                      | 17.28          | 0.77                         | 0.62          | 2.83                          | 2.49          |
| Size                                                                                                                | -21.26                     | -15.86         | -1.30                        | -1.01         | -4.92                         | -3.72         |
| <i>Panel B: Fama-MacBeth premia; <math>\bar{\gamma}</math> in %/mo, alphas in %/yr per 1 SD (NW <math>t</math>)</i> |                            |                |                              |               |                               |               |
|                                                                                                                     | Four-factor $\bar{\gamma}$ |                | Four-factor, market-adjusted |               | Beta control, market-adjusted |               |
|                                                                                                                     | Main                       | PIT            | Main                         | PIT           | Main                          | PIT           |
| Momentum                                                                                                            | 0.296 (2.91)               | 0.162 (1.76)   | 3.68 (3.41)                  | 2.22 (2.26)   | 3.68 (3.62)                   | 2.30 (2.51)   |
| Reversal                                                                                                            | 0.050 (0.51)               | 0.031 (0.36)   | 1.28 (1.05)                  | 1.08 (0.96)   | 1.04 (0.86)                   | 0.88 (0.79)   |
| Total volatility                                                                                                    | 0.493 (3.07)               | 0.365 (2.75)   | 2.52 (1.63)                  | 1.17 (0.93)   | 3.59 (3.05)                   | 1.94 (1.79)   |
| Size                                                                                                                | -0.282 (-3.85)             | -0.228 (-3.01) | -3.03 (-3.03)                | -2.39 (-2.38) | -2.69 (-2.66)                 | -2.18 (-2.21) |

Table 15 reports the results. The size spread shrinks from  $-21.26\%$  to  $-15.86\%$  a year, a reduction of 25.4%, and its Fama-MacBeth premium remains significant ( $t = -3.01$ ). Under the pre-specified rule the size diagnosis is only partially supported, and only just; the remaining negative premium is consistent with the missing exits, but this check cannot establish that. The residual volatility premium falls from 3.59% a year ( $t = 3.05$ ) to 1.94% ( $t = 1.79$ ), no longer significant, which supports the diagnosis; the volatility spread falls from 23.16% to 17.28% a year. Momentum weakens too: its decile spread falls from 7.31% to 2.43% a year ( $t = 0.41$ ), its unadjusted premium from 0.296% to 0.162% a month ( $t = 1.76$ ), and its market-adjusted premium from 3.68% ( $t = 3.41$ ) to 2.22% ( $t = 2.26$ ), still significant at 5%. End-of-sample selection accounts for a substantial part of momentum’s full-sample strength.

## 5.8 Non-winsorised factors

The final extension repeats the decile sorts and the four-factor Fama-MacBeth regression with momentum, reversal and volatility left unwinsorised. The decile spreads are unchanged, because winsorisation only ties the most extreme 1% of values at the cut-off and those stocks stay in the extreme deciles. Momentum’s Fama-MacBeth premium is stronger without winsorisation (0.332% per month,  $t = 3.29$ , against 0.296%,  $t = 2.91$ ), and its market-adjusted premium rises to 4.10% a year ( $t = 3.64$ ). Winsorisation is therefore conservative for momentum rather than flattering, and the momentum premium is not an artefact of trimming extreme values.

## 6 Limitations

**Survivorship bias (the main limitation).** A company that was small in 2012 and is among the 200 largest in 2026 must have grown enormously, and companies that shrank, were acquired or failed are absent. This is the most likely explanation for the strongly negative size premium, which says nothing about size; the same bias favours high-variance stocks and is strongest near the selection date. The point-in-time filter cannot restore firms that left the index; only a survivorship-free database with delisting returns (for example CRSP) would remove the bias fully.

**Single-episode dependence and multiple testing.** The raw significance of momentum and volatility is concentrated in the 44 months from January 2023. Momentum’s market-adjusted premium survives outside that period, but at  $t = 2.49$  it falls short of the  $t > 3$  hurdle proposed by Harvey et al. (2016). Conclusions rest on agreement across specifications, not on any single  $t$ -statistic.

**Risk-model asymmetry.** The multi-factor alphas use factors built from the survivorship-free CRSP universe to explain portfolios drawn from a universe that is not, so an alpha can measure the selection as much as any mispricing, as the volatility result shows. The regressions also omit value, profitability and investment characteristics, and the volatility factor is total rather than idiosyncratic volatility.

**Post hoc extensions.** The drawdown, multi-factor, point-in-time and non-winsorised analyses were added after the original results were known. Their specifications were committed before they were computed, which rules out choosing among variants after seeing the output, but they add to the number of specifications reported, and the questions were chosen with knowledge of where the original results looked weak.

**Look-ahead in auxiliary data.** Market capitalisation uses the current share count for every month, and GICS sectors are current labels applied historically. Neither affects the return timing, but each uses information not available at the time. The historical membership lists come from an open-source compilation, so individual dates may be wrong by days.

**Power.** About 15 years of monthly data and 200 stocks give roughly 20 stocks per decile, so decile portfolios are noisy, and the out-of-sample period has only 70 months.

**Costs and portfolio construction.** Costs are a flat 10 basis points per unit of turnover, with no market impact, borrowing cost for the short leg or within-month weight drift. Portfolios are equal-weighted, which overweights the smaller names relative to a capacity-aware implementation, and they are not beta-neutral by construction: market exposure is removed only after the fact by regression.

## 7 Conclusion

Among the 200 largest current S&P 500 constituents, 12-1 momentum is the only positive premium that remains significant both once market exposure is removed and under point-in-time index membership (market-adjusted Fama-MacBeth  $t = 3.41$ ; 2.26 point-in-time, where the unadjusted premium is no longer significant,  $t = 1.76$ ). It is the standard momentum factor (UMD loading  $t = 8.83$ ), not a separate source of return, and its decile portfolio lost 56.3% between 2016 and 2023. The strongest raw result, volatility, is market beta under the CAPM, and its large multi-factor alpha has the signature of ex post selection rather than of a premium an investor could have captured. Short-term reversal shows no reliable premium after costs. The strongly negative size premium and the residual volatility premium are attributed to survivorship bias; the point-in-time filter removes the significance of the second and shrinks the first by a quarter, which partially confirms the diagnosis.

Qualifications remain: rank ICs are insignificant for momentum and volatility, so their predic-

tive content sits in the tails; no premium outside 2023 to 2026 clears the  $t > 3$  hurdle; and firms that left the index are missing. The main value of the study is its process: a single, verified timing convention, three evaluation lenses whose disagreements are explained, risk adjustment that separates a premium from market exposure and from sample selection, and robustness extensions specified before they were computed. The natural next step is a survivorship-free universe with delisting returns.

## Reproducibility

All code is in the project repository and runs in this order: `fetch_universe.py`, `fetch_market_caps.py`, `fetch_prices.py`, `fetch_market.py`, `fetch_french.py`, `fetch_sp500_history.py`, `sanity_check_prices.py`, the five `compute_*.py` factor and beta scripts, `check_factor_distributions.py`, `winsorize_factors.py`, `decile_backtest.py`, `fama_macbeth.py`, `validation_costs.py`, `extensions.py` (sector-neutral factors), `risk_adjustment.py`, `information_coefficient.py`, then the robustness extensions `drawdowns.py`, `multifactor_alphas.py`, `point_in_time.py` and `raw_factors.py` (specified in advance in `robustness_spec.py`), and finally `make_figures.py` and `make_tables.py`. Every table and figure in this paper is generated directly from the saved outputs, so no tabulated number is transcribed by hand.

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